

L^AT_EX

*It is a language that produces
documents.*

TeX (*Greek tecnologia*)

– computer program created by

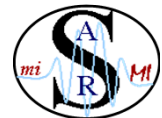
Dr. Donald E. Knuth (*Stanford University*) – 1982



LaTeX

– a macro package based on *TeX* with predefined layouts

Dr. Leslie Lamport (*Microsoft Research*) - 1980s



Some advantages of L^AT_EX

- Mathematical formulae easily described:

$$\int \int_0^\infty \frac{1}{x^2} dx \sum_{i=0}^{10} \sin^{-1}(i) \sqrt{\frac{e^x}{x\sqrt{e}}} dx$$

- Output comparable (equivalent) to that of published books;

.

Pronunciation Guide

Latex: pronounced *lay-tek*s



Pronunciation Guide

Latex: pronounced *lay-tek*s

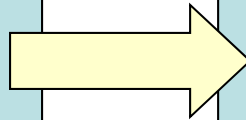


L^AT_EX: pronounced *lay-tech* (*tech* as in technical)

```
\documentclass{article}
\begin{document}

\emph{Long Live \LaTeX}

Note the emphasis!
\end{document}
```



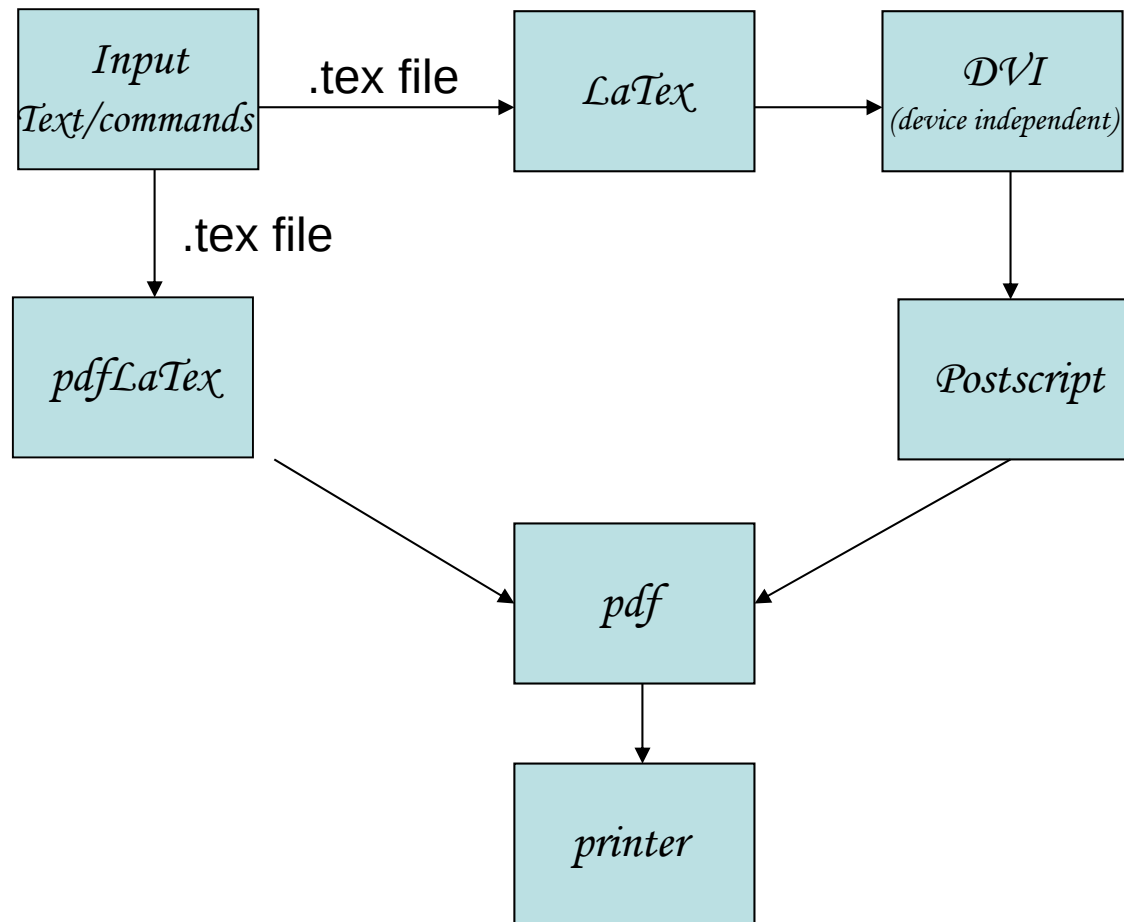
Long Live L^AT_EX
Note the emphasis!

$$\int \int_0^\infty \frac{1}{x^2} \, \mathrm{d}x \sum_{i=0}^{10} \sin^{-1}(i) \sqrt{\frac{e^x}{x\sqrt{e}}} \, \mathrm{d}x$$

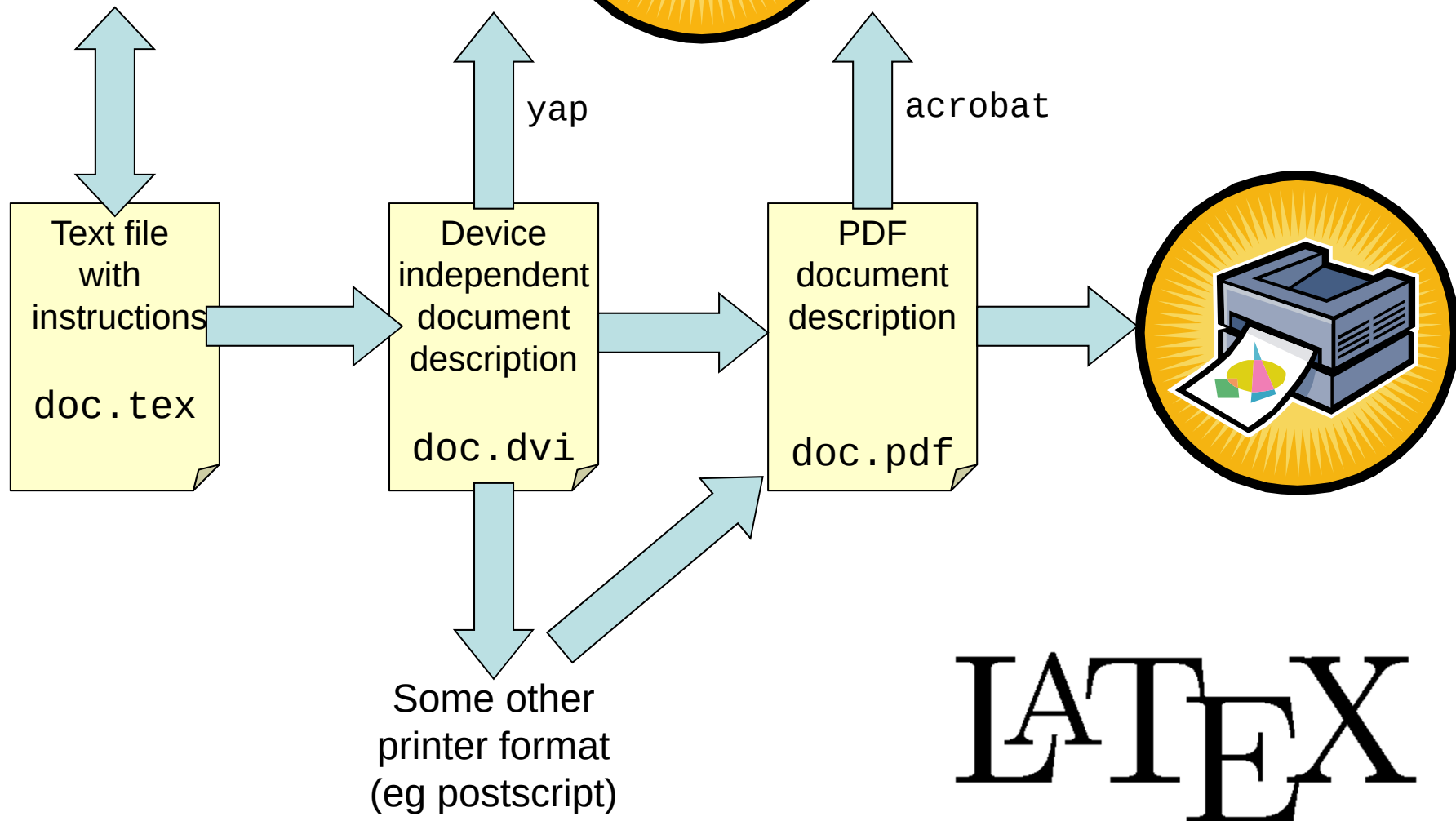
MS Word vs. LaTeX

- *Focus on content then the appearance*
- *Best for writing lot of Math*
- *References - Bibliography*
- *Manage large documents*
- *Professional output quality*
- *Predefined layouts*
- *Portable – any OS – hardware etc*
- *Creative*
- *Sizes for each part of document fixed*
- *Templates available*
- *FREE*

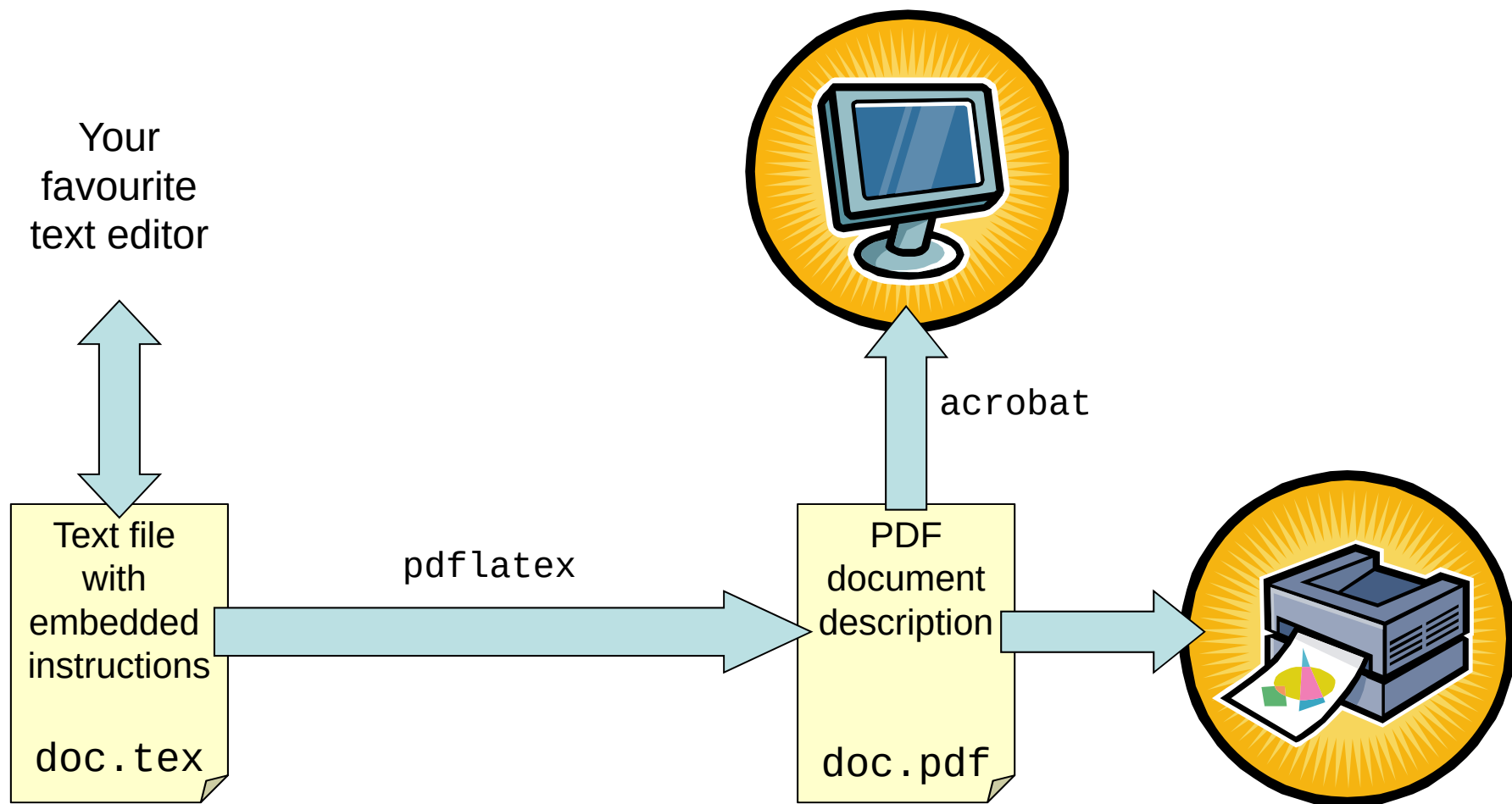
LaTeX Production Chain



Your
favourite
text editor



L^AT_EX



pdfT_EX

What to Download

- MiKTeX is a good Windows L^AT_EX implementation;
- Texniccenter is a good editor for L^AT_EX with a full IDE for document writing;
- TeXPoint is perfect for including L^AT_EX stuff in PowerPoint presentations.
- Adobe Reader for pdf viewing

Installation Method

- Explained on board
- Please include the Miktex path in the texniccenter installation procedure.

Your TeX-distribution supports the generation of PDF-files

Enter the full path of the PDF-viewer to use:

`C:\miktex\bin\x64\miktex-texworks.exe`



[optional]

Enter the command line option that prevents the viewer from starting multiple instances:

[optional]

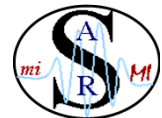
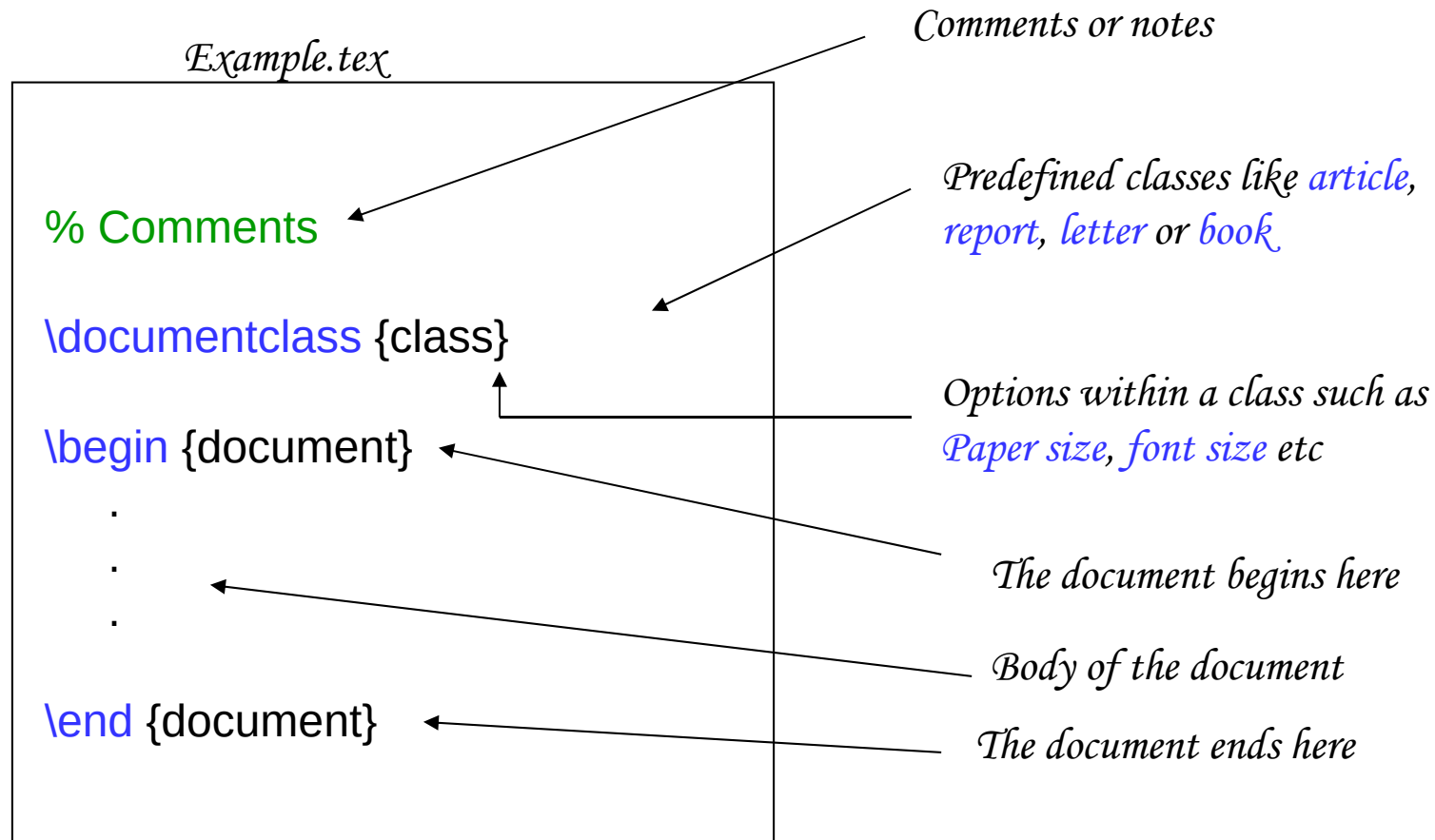
Enter the command line option, that causes the viewer to display a paragraph belonging to a given source line. Use the place holders:

‘Coding’ Basics

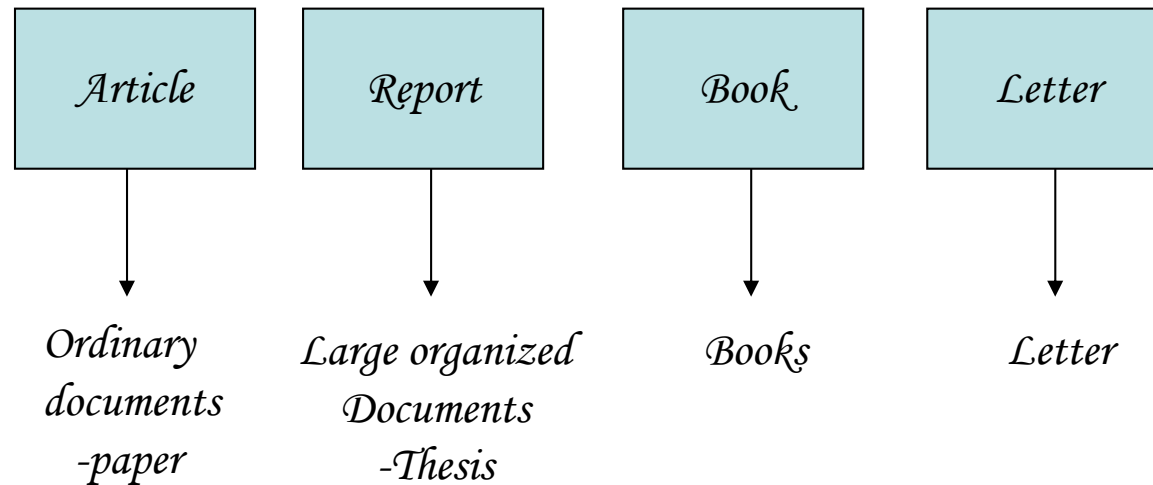
- $\text{\LaTeX}$ commands start with a backslash \
- Parameters are given in curly brackets { }
- Environments (blocks with a certain type of content) are of the form:

```
\begin{environment_type}  
environment content  
\end{environment_type}
```

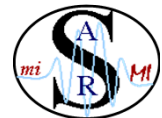
Writing a document in Latex



LaTeX Classes



*Many new class files can be created modifying the above
4 base class files
(Ex: IEEEtran.cls – derived from article.cls)*



The Simplest Document

```
\documentclass{report}
```

```
\begin{document}
```

```
\end{document}
```

Hello World!

```
\documentclass{report}  
\begin{document}  
Hello  
World!  
  
Bye!  
\end{document}
```

Hello World!

```
\documentclass{report}  
\begin{document}  
Hello  
World!  
  
Bye!  
\end{document}
```

Hello World!
Bye!

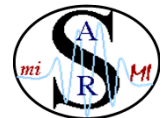
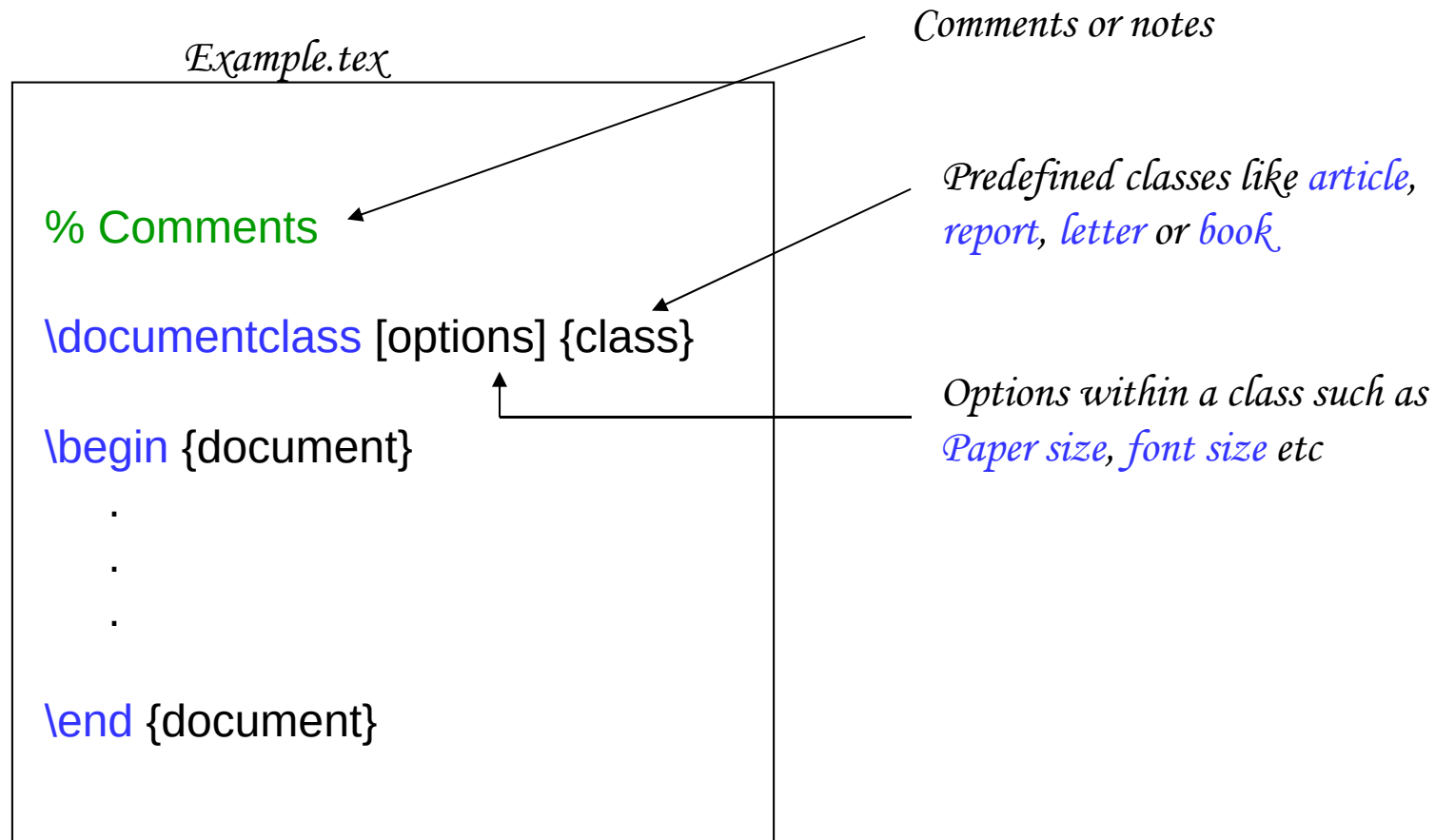
Hello World

```
\documentclass{report}  
\begin{document}  
Hello  
World!  
  
Bye!  
\end{document}
```

- Newlines in the source are ignored.
- Skipping a line in the source starts a new paragraph

Hello World!
Bye!

Writing a document in Latex



Writing a document in Latex

Example.tex

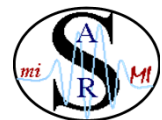
```
\documentclass [12pt,a4paper] {article}
```

```
\begin {document}
```

```
.  
.   
.
```

```
\end {document}
```

*Options within a class such as
font size, paper size etc*



Useful Document Classes

- **Article:** Ideal for a short paper (divided into sections, subsections, etc). Typeset to be printed double-sided.
- **Book:** Class to be used to typeset a book (chapters, sections, etc). Typeset to be printed double-sided.
- **Report:** (Almost) identical to the book class but for single-sided printing.
- Other classes include **letter**, **slides**, etc.

Organising a Document

- A document is split into logical parts:
 - A title
 - An abstract
 - A number of parts
 - A number of chapters in each part
 - A number of sections in each chapter
 - A number of subsections in each section
 - A number of subsubsections in each subsection
- Availability of a logical part depends on the document class (eg no chapters in a paper).
- Typesetting of the sections may vary depending on document class.

The Title

```
\documentclass{article}  
\begin{document}  
  
\title{About College of EME}  
\author{Dean College of EME}  
\date{28th March 17}  
\maketitle  
  
\end{document}
```

The Title

```
\documentclass{article}  
\begin{document}  
  
\title{About College of EME}  
\author{Dean College of EME}  
\date{28th March 17}  
\maketitle  
  
\end{document}
```

About College of EME

Dean College of EME

28th March 17

The Abstract

- Available in article and report class.
- Used to give an overview of the content of the document.
- Is usually typeset with wider margins than the main text.
- Specified using the abstract environment:

```
\begin{abstract}
```

```
...
```

```
\end{abstract}
```

Abstract Example

```
\documentclass{article}  
  \begin{document}  
    \begin{abstract}  
      Your abstract goes here... ..  
    \end{abstract} ...  
  \end{document}
```

Logical Sections

- To start a named part, chapter, section, subsection or subsubsection use the command:

`\type{name}`

where `type` is one of `part`, `chapter`, `section`, `subsection` and `subsubsection`.

- This will automatically create the title and numbering of the section.
- Any text after the command will appear inside the section named.
- Parts and chapters are only available in the report and book class.

Logical Sections

```
\documentclass{report}  
\begin{document}  
\chapter{Basics}  
  
\section{First of all}  
Initially, we think.  
  
\subsection{Sit Down}  
Find a chair and sit down.  
  
\subsection{Think}  
Think about the chair.  
  
\end{document}
```

Logical Sections and their Presence

Command	Level	Comment
<code>\part{"part"}</code>	-1	not in letters
<code>\chapter{"chapter"}</code>	0	only books and reports
<code>\section{"section"}</code>	1	not in letters
<code>\subsection{"subsection"}</code>	2	not in letters
<code>\subsubsection{"subsubse ction"}</code>	3	not in letters
<code>\paragraph{"paragraph"}</code>	4	not in letters
<code>\subparagraph{"subparagr aph"}</code>	5	not in letters

Logical Sections

```
\documentclass{report}
\begin{document}
\chapter{Basics}

\section{First of all \ldots}
Initially, we think.

\subsection{Sit Down}
Find a chair and sit down.

\subsection{Think}
Think about the chair.

\end{document}
```

Chapter 1

Basics

1.1 First of all ...

Initially, we think.

1.1.1 Sit Down

Find a chair and sit down.

1.1.2 Think

Think about the chair.

Tables

- To draw up tabular data, use the `tabular` environment.
- An extra parameter gives the information about the column layout.
- Separate lines using `\\` and columns using the ampersand (`&`) symbol.
- `\hline` draws a horizontal line.

Tables

- To draw up tabular data, use the `tabular` environment.
- An extra parameter gives the information about the column layout.
- Separate lines using `\\` and using the ampersand (`&`) to separate cells.
- `\hline` draws a horizontal line.

eg `{|l|c|r|}` results in three columns separated by vertical lines, a left aligned column, another vertical line, a centred column, and a right aligned column, and a vertical line.

Eurovision Singers

```
\begin{tabular}{|l|l|l|}  
\hline  
Name & Pos & Pnts \\ \hline  
Ali & 3rd & 5 \\  
Hassan & 2nd & 10 \\  
Saeed & 1st & 15 \\ \hline  
\end{tabular}
```

Eurovision Singers

```
\begin{tabular}{|c|c|c|}  
\hline  
Name & Pos & Pnts \\\hline  
Ali & 3rd & 5 \\\hline  
Hassan & 2nd & 10 \\\hline  
Saeed & 1st & 15 \\\hline  
\end{tabular}
```

Name	Pos	Pnts
Ali	3rd	5
Hassan	2nd	10
Saeed	1st	15

Vertical Bars are Optional

```
\begin{tabular}{l c r }  
  1 & 2 & 3 \\  
  4 & 5 & 6 \\  
  7 & 8 & 9 \\  
\end{tabular}
```

1	2	3
4	5	6
7	8	9

Expanding upon that by including some vertical lines:

```
\begin{tabular}{l | c | r }  
  1 & 2 & 3 \\  
  4 & 5 & 6 \\  
  7 & 8 & 9 \\  
\end{tabular}
```

1	2	3
4	5	6
7	8	9

Figures

- The `figure` environment is used to include a *floating* figure in the text.
- A caption can be added to both using the `\caption{}` command.
- The two environments are identical except for the caption title, and whether they appear in a list of figures, or the list of tables.

Figures and Tables

- The `figure` environment is used to include a *floating* figure in the text.
- A caption can be added to both using the `\caption` command.
- The text of the caption can appear in a list of figures or tables.

Floating objects do not have a fixed position in the body of the text, but may be moved around by `\float` to improve the layout

Figures and Tables

```
\begin{figure}  
Here include whatever you  
want in the figure.
```

```
\caption{A demo figure}  
\end{figure}
```

Note that the figures and
tables may move \ldots

Figures and Tables

```
\begin{figure}  
Here include whatever you  
want in the figure
```

```
\caption{A demo fi  
\end{figure}
```

Note that the figure
tables may move \l

Here include whatever you want in the figure.

Figure 1: A demo figure

Here include whatever you want in the table.

Table 1: A demo table

Any other text ...

Note that the figures and tables may move ...

Figures and Tables

L^AT_EX

```
\begin{figure}  
Here include whatever you  
want in the figure
```

```
\caption{A demo fi  
\end{figure}
```

Note that the figu
tables may move \l

keeps count of the figure and
table numbering
automatically.

Here include whatever you want in the figure

Figure 1: A demo figure

Here include whatever you want in the table.

Table 1: A demo table

Any other text ...

Note that the figures and tables may move ...

Figures and Tables

```
\begin{figure}  
Here include whatever you  
want in the figure.  
\includegraphics[scale=0.7]{Images/sysModelChannDes}  
\caption{A demo figure}  
\end{figure}
```

Any other text \lo

```
\begin{table}  
Here include whate  
want in the table.
```

```
\caption{A demo ta  
\table}
```

Note that the figu
tables may move \l

Here include whatever you want in the figure.

Figure 1: A demo figure

Here include whatever you want in the table.

Table 1: A demo table

Any other text ...

Note that the figures and tables may move ...



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Formats that work with LaTeX (`dvi` mode, using `dvips`):

- `eps`

Formats that work with LaTeX (`dvi` mode, using `dvipdfm` (`x`)):

- `pdf`
- `png`
- `jpeg`
- `eps`

Formats that work with pdfLaTeX (`pdf` mode):

- `pdf`
- `png`
- `jpeg`

Tables of Contents

- To add a table of contents, with parts, chapters, sections, etc use the command `\tableofcontents`.
- You may also include a list of figures and a list of tables using `\listoffigures` and `\listoftables`.

Tables of Contents

```
\title{A Quick \LaTeX\ Primer}  
\author{Gordon J. Pace}  
\date{}  
\maketitle  
  
\tableofcontents
```

Tables of Contents

```
\title{A Quick \LaTeX\ Primer}  
\author{Gordon J. Pace}  
\date{}  
\maketitle  
  
\tableofcontents
```

A Quick $\text{\LaTeX}$ Primer

Gordon J. Pace

Contents

1	Introduction	3
1.1	Motivation	3
1.2	What is $\text{\LaTeX}$?	3
1.3	Trivia	3
2	Producing Output	3
3	The first few steps	4
3.1	Starting a document	4
3.2	Paragraphs and line breaks	4
3.3	Titles	5
4	Frills	5
4.1	Typefaces	5
4.2	Type Size	6
4.3	Verbatim Environment	6

Ta

CAUTION

$\text{\LaTeX}$ creates an auxiliary file with all page numbers when run through a file. Run $\text{\LaTeX}$ twice to ensure that they match

```
\title{A Quick \LaTeX Primer}
\author{Gordon J. Pace}
\date{}
\maketitle
```

```
\tableofcontents
```

A Quick $\text{\LaTeX}$ Primer

Gordon J. Pace

Contents

1	Introduction	3
1.1	Motivation	3
1.2	What is $\text{\LaTeX}$?	3
1.3	Trivia	3
2	Producing Output	3
3	The first few steps	4
3.1	Starting a document	4
3.2	Paragraphs and line breaks	4
3.3	Titles	5
4	Frills	5
4.1	Typefaces	5
4.2	Type Size	6
4.3	Verbatim Environment	6

Cross References

- Quite regularly, you will want to refer to section 4.2.1.4 from section 7.6.4.2.
- But when you add a section before section 4.2, you will have to revise the numbering...
- $\text{\LaTeX}$ provides a way of naming sections, chapters, figures and tables to allow references updated automatically.

Cross References

- Use `\label{label name}` to name a numbered object (figure, section, chapter, etc). It may appear within the `\caption`, `\section`, etc parameter or just after it.
- Use `\ref{label name}` to insert the number of the object named using `\label` command.
- Use `\pageref{label name}` to insert the page number where the named object appears.
- References may be forward or backward ones.

Cross References

```
\section{Introduction}

... more details can be
found in section \ref{s:proof}
(in particular look at figure
\ref{f:proof} on page
\pageref{f:proof})...

\section{Proofs and Roofs}
...

\subsection{Proof \label{s:proof}}

\begin{figure}
...
\caption{Proof outline
  \label{f:proof}}
\end{figure}
...
```

Cross References

```
\section{Introduction}

... more details can be
found in section \ref{s:proof}
(in particular look at figure
\ref{f:proof} on page
\pageref{f:proof})...

\section{Proofs and Roofs}
...

\subsection{Proof \label{f:proof}}

\begin{figure}
...
\caption{Proof outline
\label{f:proof}}
\end{figure}
...
```

1 Introduction

...more details can be found in section 3.2 (in particular look at figure 8 on page 32).

...

3 Proofs and Roofs

...

3.2 Proof

...

...

Figure 3: Proof outline

...

Cr

Using labels starting with f:
(for figures), s: (for
sections), c: (for chapters)
is not obligatory but will
help you remember.

```
\section{Introduction}
```

```
... more details can  
found in section \ref{f:proof}  
(in particular look at  
\ref{f:proof} on page  
\pageref{f:proof})...
```

```
\section{Proofs and Roofs}  
...
```

```
\subsection{Proof \label{f:proof}}
```

```
\begin{figure}
```

```
...
```

```
\caption{Proof outline  
 \label{f:proof}}
```

```
\end{figure}
```

```
...
```

1 Introduction

...more details can be found in section 3.2 (in particular look at figure 8 on page 32).

...

3 Proofs and Roofs

...

3.2 Proof

...

...

Figure 3: Proof outline

...

C

CAUTION

As with tables of contents, $\text{\LaTeX}$ uses an auxiliary file with all references. Run $\text{\LaTeX}$ twice to ensure that references match

```
\section{Introduction}
```

```
... more details can be found in section 3.2 (in particular look at figure 8 on page 32).  
\ref{f:proof} on page  
\pageref{f:proof})...
```

```
\section{Proofs and Roofs}  
...
```

```
\subsection{Proof \label{f:proof}}
```

```
\begin{figure}
```

```
...
```

```
\caption{Proof outline  
 \label{f:proof}}
```

```
\end{figure}
```

```
...
```

1 Introduction

... more details can be found in section 3.2 (in particular look at figure 8 on page 32).
...

3 Proofs and Roofs

...

3.2 Proof

...

...

Figure 3: Proof outline

...

Organising a Large Document

- Writing a large document can make access to different parts unwieldy.
- $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ has two commands to import files from one another: `\input{file}`, `\include{file}`.
- Both include the text in the file given as parameter as-is into the main document.
- They are identical except that `\include` starts a new page automatically.

Organising a Large Document

- Writing a large document can make access to different parts unwieldy.
- $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ has two commands to import files from one another: `\input{file}`, `\include{file}`.
- Both include the text in the parameter as-is into the document.
- They are identical except `\include` starts a new page automatically.

Note that although the files are assumed to be $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ files, the `.tex` extension is left out.

Typical Document Organisation

main.tex

```
% Document: FYP Chicken-Sort
% Date:      1/9/2005
% Author:    Cikku Flieles

\documentstyle{report}
\begin{document}
\input{frontpage}

% short introduction
\input{introduction}

% The algorithm
\input{algorithm}

% Correctness proof
\input{proof}

% Conclusions
\input{conclusions}
\end{document}
```

Typical Document Organisation

main.tex

```
% Document: FYP Chicken-Sort
% Date:      1/9/2005
% Author:    Cikku Flieles

\documentstyle{report}
\begin{document}
\input{frontpage}

% short introduction
\input{introduction}

% The algorithm
\input{algorithm}

% Correctness proof
\input{proof}

% Conclusions
\input{conclusions}
\end{document}
```

proof.tex

```
\chapter{The Proof of
          Correctness}

\section{Proof Outline}

We prove the correctness of
chicken sort using a technique
based on the Socratic method
of discourse.

Let us start by assuming that
Plato knew about this
proof...
```

Images

- Images can be added anywhere in a document (not just as a figure)
- To include an image, use the command `\includegraphics{filename}`
- Various formats are allowed, including gif, jpg, pdf (when using pdftex), etc.

Resizing Images

- Images can be resized using the

```
\resizebox{width}{height}  
{object to resize}
```

- Width and height can be set in various units: cm, mm, in, textwidth (the size of the page less the margins).
- Use ! as the width or height to scale it according to the other given size.

Il-Ahwa x'Figure!

```
\begin{figure}  
\resizebox{0.8textwidth}{!}  
  {\includegraphics{girlinlatex.gif}}  
\caption{Il-Ahwa x'Figure}  
\end{figure}
```

Il-Ahwa x'Figure!

```
\begin{figure}  
\resizebox{0.8textwidth}{!}  
  {\includegraphics{girlinlatex.gif}}  
\caption{Il-Ahwa x'Figure}  
\end{figure}
```

L_{at}_{ex}T_EX

Figure 1: Il-Ahwa x'Figure

Footnotes

- To add a footnote, use the:

`\footnote{footnote text}`

- A footnote mark (such as ¹ or ²) appears where the command is given, and the footnote at the bottom of the page.
- $\text{\LaTeX}$ takes care of the rest.

Unordered Lists

- The `itemize` environment creates an unordered list (like this one).
- Items are started using the `\item` command.
- You can have nested lists.

Ordered Lists

1. The `enumerate` environment creates an unordered list (like this one).
2. Items are also started using the `\item` command.
3. Nested ordered lists can be used.

Description Lists

The environment: The `\description` environment creates a description list (like this one).

Items: Items are started using the `\item` command, with the title given as an optional parameter (in square brackets) just after the command.

Elephants and Giraffes

```
\begin{description}
\item[Elephants:] Elephants can be
    typeset using the \verb+\elephant+
    command.
\item[Giraffes:] Two points should be
    kept in mind:
    \begin{itemize}
        \item \LaTeX\ offers no support
            for giraffes.
        \item Neither does Word.
    \end{itemize}
\end{description}
```

Elephants and Giraffes

```
\begin{description}
```

```
\item[Elephants:] Elephants can be
```

```
\item
```

Elephants: Elephants can be typeset
using the `\elephant` command.

Giraffes: Two points should be kept in
mind:

- L^AT_EX offers no support for giraffes.
- Neither does Word.

```
\end
```

Emphasis and Boldface

- *Emphasised* text is produced using the `\emph{text}` command.
- **Boldface** is obtained through the use of `\textbf{text}`.
- Rarely used are sans-serif `\textsf{text}`, small caps `\textsc{text}` and typewritten `\texttt{text}`.

Bold Elephants in Occam

`\textbf{Definition:}` An `\emph{elephant}` can be defined to be a giraffe who should go on a diet.

`\textbf{Definition:}` A `\emph{giraffe}` is nothing but an anorexic elephant.

`\textsf{Occam}` implements elephants very efficiently. Note that `\textsf{Occam}` is not written `\textsc{Occam}` or `\texttt{Occam}`!

Bold Elephants in Occam

```
\textbf{Definition:} An \emph{elephant}
```

```
can  
go o
```

```
\tex  
is n
```

```
\tex  
effi  
not  
\tex
```

Definition: An *elephant* can be defined to be a giraffe who should go on a diet.

Definition: A *giraffe* is nothing but an anorexic elephant.

Occam implements elephants very efficiently. Note that **Occam** is not written OCCAM or Occam!

Some Symbols

- To illustrate the rest of this part, we will introduce some mathematical symbols. More can be found on the last few slides.
- All mathematics must appear in maths mode – but more about this in a moment...
- The following symbols can be produced using the commands: `\leq`, `\times`, `\pi`, `\infty`.

$$\leq \quad \times \quad \pi \quad \infty$$

More Symbols

Some symbols are used to combine other mathematical expressions:

- Powers:

$$e^{i\pi} = -1$$

$$e^{i\pi} = -1$$

- Subscripts:

$$a_{n+1} = 2 \times a_n$$

$$a_{n+1} = 2 \times a_n$$

- Fractions:

$$\frac{x^\pi}{a_{25}}$$

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- Summation:

$$\sum_{i=0}^{\infty} a^i$$

$$\sum_{i=0}^{\infty} a^i$$

Inline Mathematics

- If a mathematical expression appears in a line of normal text, use a dollar symbol \$ to start and to end the mathematics.
- This ensures that the lines are kept as narrow as possible to avoid L^AT_EX having to change line spacing drastically.

Inline Mathematics

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- This ensures that the lines are kept as low as possible to avoid L^AT_EX having to change line spacing.

It has already been shown that $a_{n+1} = 2 \times a_n$. We can thus conclude that $\frac{a_n}{a_0} = 2^n$.

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Large Formulae

- If the mathematical formulae are to appear on a separate line, start the mathematics using `\[`, and end it with `\]`.

It has already been shown that $a_{n+1} = 2 \times a_n$. We can thus conclude that:

$$\left[\frac{a_n}{a_0} = 2^n \right]$$

Note the difference from inline mode: $\frac{a_n}{a_0} = 2^n$.

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Note the difference from inline mode: `\frac{a_n}{a_0} = 2^n`.

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$$\frac{a_n}{a_0} = 2^n$$

Note the difference from inline mode: $\frac{a_n}{a_0} = 2^n$.

Equations

- Quite regularly, one needs equations with aligned equality signs. The environment `eqnarray` is used for this.
- Separate lines using `\\`, and the left hand side, equality, and right hand side of the equations using `&`.

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Equations

- Quite regularly, one needs equations with aligned equality signs. The environment `eqnarray` is used for this
- Separate lines using `&=` on the left side, equality, and `×` on the right side, equations using `\times`

Note that the equations are numbered. If you don't want this feature, use the `eqnarray*` environment instead

```
\begin{eqnarray}
a_0 &=& 1 \\
a_{n+1} &=& 2\times a_n
\end{eqnarray}
```

$$\begin{array}{rcl} a_0 & = & 1 \\ a_{n+1} & = & 2 \times a_n \end{array} \begin{array}{l} (1) \\ (2) \end{array}$$

Some Notes ...

- Spacing is ignored in math mode – $x y$ gives the same result as xy .
- If you need to escape from math mode to include normal text (eg for the name of a function) use the command `\mbox{text}` (eg $e^n + \mbox{fibonacci}(n+1)$)
- Never use math mode to emphasise text – because different gives *different*, not *different*!

Some Mathematical Symbols

$\aleph$	<code>\aleph</code>	$'$	<code>\prime</code>	$\forall$	<code>\forall</code>	<code>\forall</code>	<code>\forall</code>
$\hbar$	<code>\hbar</code>	$\emptyset$	<code>\emptyset</code>	$\exists$	<code>\exists</code>	$\exists$	<code>\exists</code>
$\imath$	<code>\imath</code>	∇	<code>\nabla</code>	$\neg$	<code>\neg</code>	$\neg$	<code>\neg</code>
$\jmath$	<code>\jmath</code>	$\surd$	<code>\surd</code>	$\flat$	<code>\flat</code>	$\flat$	<code>\flat</code>
ℓ	<code>\ell</code>	$\top$	<code>\top</code>	$\natural$	<code>\natural</code>	$\natural$	<code>\natural</code>
$\wp$	<code>\wp</code>	$\bot$	<code>\bot</code>	$\sharp$	<code>\sharp</code>	$\sharp$	<code>\sharp</code>
$\Re$	<code>\Re</code>	$\parallel$	<code>\parallel</code>	$\clubsuit$	<code>\clubsuit</code>	$\clubsuit$	<code>\clubsuit</code>
$\Im$	<code>\Im</code>	$\angle$	<code>\angle</code>	$\diamondsuit$	<code>\diamondsuit</code>	$\diamondsuit$	<code>\diamondsuit</code>
∂	<code>\partial</code>	$\triangle$	<code>\triangle</code>	$\heartsuit$	<code>\heartsuit</code>	$\heartsuit$	<code>\heartsuit</code>
∞	<code>\infty</code>	$\backslash$	<code>\backslash</code>	$\spadesuit$	<code>\spadesuit</code>	$\spadesuit$	<code>\spadesuit</code>

Binary Operators

$\pm$	<code>\pm</code>	$\cap$	<code>\cap</code>	$\vee$	<code>\vee</code>
$\mp$	<code>\mp</code>	$\cup$	<code>\cup</code>	$\wedge$	<code>\wedge</code>
$\setminus$	<code>\setminus</code>	$\oplus$	<code>\oplus</code>	$\oplus$	<code>\oplus</code>
$\cdot$	<code>\cdot</code>	$\sqcap$	<code>\sqcap</code>	$\ominus$	<code>\ominus</code>
$\times$	<code>\times</code>	$\sqcup$	<code>\sqcup</code>	$\otimes$	<code>\otimes</code>
$*$	<code>\ast</code>	$\triangleleft$	<code>\triangleleft</code>	$\oslash$	<code>\oslash</code>
$\star$	<code>\star</code>	$\triangleright$	<code>\triangleright</code>	$\odot$	<code>\odot</code>
$\diamond$	<code>\diamond</code>	$\wr$	<code>\wr</code>	$\dagger$	<code>\dagger</code>
$\circ$	<code>\circ</code>	$\bigcirc$	<code>\bigcirc</code>	$\ddagger$	<code>\ddagger</code>
$\bullet$	<code>\bullet</code>	$\triangleup$	<code>\triangleup</code>	$\amalg$	<code>\amalg</code>
$\div$	<code>\div</code>	$\triangledown$	<code>\triangledown</code>		

Relational Operators

$\leq$	<code>\leq</code>	$\geq$	<code>\geq</code>	$\equiv$	<code>\equiv</code>
$\prec$	<code>\prec</code>	$\succ$	<code>\succ</code>	$\sim$	<code>\sim</code>
$\preceq$	<code>\preceq</code>	$\succeq$	<code>\succeq</code>	$\simeq$	<code>\simeq</code>
$\ll$	<code>\ll</code>	$\gg$	<code>\gg</code>	$\asymp$	<code>\asymp</code>
$\subset$	<code>\subset</code>	$\supset$	<code>\supset</code>	$\approx$	<code>\approx</code>
$\subseteq$	<code>\subseteq</code>	$\supseteq$	<code>\supseteq</code>	$\cong$	<code>\cong</code>
$\sqsubseteq$	<code>\sqsubseteq</code>	$\sqsupseteq$	<code>\sqsupseteq</code>	$\bowtie$	<code>\bowtie</code>
$\in$	<code>\in</code>	$\ni$	<code>\ni</code>	$\propto$	<code>\propto</code>
$\vdash$	<code>\vdash</code>	$\dashv$	<code>\dashv</code>	$\models$	<code>\models</code>
$\smile$	<code>\smile</code>	$\mid$	<code>\mid</code>	$\doteq$	<code>\doteq</code>
$\frown$	<code>\frown</code>	$\parallel$	<code>\parallel</code>	$\perp$	<code>\perp</code>

Some Arrows

$\leftarrow$	<code>\leftarrow</code>	$\rightarrow$	<code>\rightarrow</code>
$\longleftarrow$	<code>\longleftarrow</code>	$\longrightarrow$	<code>\longrightarrow</code>
$\Leftrightarrow$	<code>\Leftrightarrow</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\Longleftarrow$	<code>\Longleftarrow</code>	$\Longrightarrow$	<code>\Longrightarrow</code>
$\leftrightarrow$	<code>\leftrightarrow</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\longleftrightarrow$	<code>\longleftrightarrow</code>	$\Longleftrightarrow$	<code>\Longleftrightarrow</code>
$\hookleftarrow$	<code>\hookleftarrow</code>	$\hookrightarrow$	<code>\hookrightarrow</code>
$\leftharpoonup$	<code>\leftharpoonup</code>	$\rightharpoonup$	<code>\rightharpoonup</code>
$\leftharpoondown$	<code>\leftharpoondown</code>	$\rightharpoondown$	<code>\rightharpoondown</code>
$\uparrow$	<code>\uparrow</code>	$\downarrow$	<code>\downarrow</code>
$\Uparrow$	<code>\Uparrow</code>	$\Downarrow$	<code>\Downarrow</code>
$\updownarrow$	<code>\updownarrow</code>	$\Updownarrow$	<code>\Updownarrow</code>
$\nearrow$	<code>\nearrow</code>	$\nwarrow$	<code>\nwarrow</code>
$\searrow$	<code>\searrow</code>	$\swarrow$	<code>\swarrow</code>
$\mapsto$	<code>\mapsto</code>	$\longmapsto$	<code>\longmapsto</code>
$\rightleftharpoons$	<code>\rightleftharpoons</code>		

Big Symbols

 Σ `\sum` $\prod$ `\prod` $\coprod$ `\coprod` $\int$ `\int` $\oint$ `\oint` $\bigcap$ `\bigcap` $\bigcup$ `\bigcup` $\bigsqcup$ `\bigsqcup` $\bigvee$ `\bigvee` $\bigwedge$ `\bigwedge` $\bigodot$ `\bigodot` $\bigotimes$ `\bigotimes` $\bigoplus$ `\bigoplus` $\biguplus$ `\biguplus`

Greek Letters

α	<code>\alpha</code>	ι	<code>\iota</code>	ρ	<code>\rho</code>
β	<code>\beta</code>	κ	<code>\kappa</code>	σ	<code>\sigma</code>
γ	<code>\gamma</code>	λ	<code>\lambda</code>	τ	<code>\tau</code>
δ	<code>\delta</code>	μ	<code>\mu</code>	υ	<code>\upsilon</code>
ϵ	<code>\epsilon</code>	ν	<code>\nu</code>	ϕ	<code>\phi</code>
ζ	<code>\zeta</code>	ξ	<code>\xi</code>	χ	<code>\chi</code>
η	<code>\eta</code>	$\omicron$	<code>\omicron</code>	ψ	<code>\psi</code>
θ	<code>\theta</code>	π	<code>\pi</code>	ω	<code>\omega</code>

ϵ	<code>\epsilon</code>	ε	<code>\varepsilon</code>
θ	<code>\theta</code>	ϑ	<code>\vartheta</code>
π	<code>\pi</code>	ϖ	<code>\varpi</code>
ρ	<code>\rho</code>	ϱ	<code>\varrho</code>
σ	<code>\sigma</code>	ς	<code>\varsigma</code>
ϕ	<code>\phi</code>	φ	<code>\varphi</code>

Standard Functions

<code>\cos</code>	<code>cos</code>	<code>\exp</code>	<code>exp</code>
<code>\arccos</code>	<code>arccos</code>	<code>\ln</code>	<code>ln</code>
<code>\sin</code>	<code>sin</code>	<code>\log</code>	<code>log</code>
<code>\arcsin</code>	<code>arcsin</code>	<code>\gcd</code>	<code>gcd</code>
<code>\tan</code>	<code>tan</code>	<code>\min</code>	<code>min</code>
<code>\arctan</code>	<code>arctan</code>	<code>\max</code>	<code>max</code>

Simple Bibliographies

- To create the bibliography, use the `thebibliography` environment.
- Items in the bibliography are added using the `\bibitem{label}` command. The label is used to refer to the entry.
- Citing a bibliography item in the main text can be done using the `\cite{label}` or `\cite{label1, label2, ...}` command to obtain citations such as [2] or [7,4].

Citing Papers the Easy Way

```
\cite{alur:94, asarin:01} talk about timed-automata, even if  
\cite{asarin:01} treats a particular case of the general case  
appearing in \cite{alur:94}.
```

```
\begin{thebibliography}{99}
```

```
\bibitem{alur:94} R. Alur and D.L. Dill,  
  \emph{A theory of timed automata, \/}  
  Theoretical Computer Science 126:1(183--235), 1994.
```

```
\bibitem{asarin:01} E. Asarin, G. Schneider and S. Yovine,  
  \emph{On the Decidability of the Reachability Problem for  
  Planar Differential Inclusions, \/} in Lecture Notes in  
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```
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\begin{thebibliography}{  
\bibitem{  
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Theor  
  
\bibitem{  
\emph{  
Planar  
Compu  
\end{thebibliography}
```

[1, 2] talk about timed-automata, even if [2] treats a particular case of the general case Appearing in [1].

References

- [1] R. Alur and D.L. Dill, *A theory of timed automata*, Theoretical Computer Science 126:1(183–235), 1994.
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  Planar Differential Automata},  
  Computer Science 2034, Lecture Notes in Computer Science,  
  Springer-Verlag, 2001.  
\end{thebibliography}
```

The number here is used to
tell ~~L~~^A~~T~~^E~~X~~ that no more than 99
entries will appear in the
bibliography

Citing Bibliography the Easy Way

CAUTION

$\text{\LaTeX}$ creates an bibliography file with all reference information when run through a file. Run $\text{\LaTeX}$ twice to ensure that references and citations match

```
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Appearin
```

```
\begin{t  
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```

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\emph{On the Decidability of the Reachability Problem for  
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Computer Science 2034, 2001.  
\end{thebibliography}
```

The Problems with This...

- It is your responsibility to ensure that the references appear in a standard way (eg all journal papers start with the author name, have an emphasised title, etc).
- Reuse of bibliographies can be cumbersome.

Using BibT_EX

BibT_EX is an additional tool, which takes a list of references (generated by L^AT_EX), a (.bib) file with all the bibliographic entries, and generates a file with all the reference entries to be used by L^AT_EX.

Using BibT_EX.

1. Run L^AT_EX to generate the list of citations.
2. Run BibT_EX to obtain the reference details.
3. Run L^AT_EX to use the reference details.
4. Run L^AT_EX again to get references right.

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Luckily, WinEdt does this at the click of one button!

Using BibT_EX.

- Citations in the L^AT_EX file appear as before using the `\cite` command.
- At the point where you want the references to appear, use the `\bibliography{file}` command, where `filename.bib` is the name of the file with the BibT_EX entries.

Your BibT_EX Will Look Like This ...

```
@article{alur:94,  
  author="R. Alur and D.L. Dill",  
  title="A theory of timed automata",  
  journal="Theoretical Computer Science",  
  year=1994,  
  volume=126,  
  pages="183- -235"  
}  
  
@Book{bird:88,  
  author = "R. Bird and P. Wadler",  
  title = "Introduction to Functional Programming",  
  publisher = "Prentice Hall International, New York",  
  year = 1988  
}
```

Your BibT_EX Will Look Like This ...

```
@article{alur:94,  
  author="R. Alur and D.  
  title="A theory of t  
  journal="Theoretical  
  year=1994,  
  volume=126,  
  pages="183--235"  
}
```

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  year = 1988  
}
```

You can either download these entries, type them in yourself, or use a bibliography database tool (eg JabRef) to manage them

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@article{alur:94,  
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  journal="Theoretical Computer Science",  
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  author = "R. Bird and",  
  title = "Introduction to the Theory of Computation",  
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  year = 1988  
}
```

BibT_EX uses the entry type (article, book, etc) to ensure a standard formatting. You don't need to worry any longer whether it's book or article titles that should appear in italics.

Using BibT_EX.

- Using BibT_EX will only show the items in the bibliography referred to in the text. Use the `\nocite{*}` command in the L^AT_EX file to show all the items in the bibliography.
- If you prefer citations of the form [Alu94] instead of [1], add the command `\bibliographystyle{alpha}` before the bibliography in the L^AT_EX file.

Quick and Dirty Bibliographies, or Bib_TE_X?

- If you're writing a quick document, with references you will use only once, don't bother with Bib_TE_X.
- However, in that case, make sure that your bibliographic entries appear in a standard way, and contain all the necessary information.
- For a bigger project, even an FYP, I would start building a Bib_TE_X database early on to reduce work later on.

Include New Packages in Latex

- Install the packages
 - Manually
 - Download
 - Unzip
 - go to cmd and folder where unzipped
 - run “latex package.ins”
 - You will get “package.sty” and other files
 - Copy package.sty to the folder where tex file is located.

Include New Packages in Latex

- Automatically (Using e.g. Miktex package Manager)

Go forth and
L^AT_EXify